

Mathematical proof - Set 1 - Solutions

Specification Prf1-Prf4.

Answer key

Question	Answer	Topic
1	E	Mathematical proof
2	D	Mathematical proof
3	A	Mathematical proof
4	E	Mathematical proof
5	D	Mathematical proof
6	D	Mathematical proof
7	B	Mathematical proof
8	B	Mathematical proof
9	A	Mathematical proof
10	B	Mathematical proof
11	A	Mathematical proof
12	A	Mathematical proof

Worked solutions

Question 1 PRF-01 · Mathematical proof · spec Prf1

Consider the claim: **for every positive integer n , $n^2 + n$ is even**. Which of the following is a complete and valid proof?

Answer: E

The factorised argument covers every n at once: $n(n + 1)$ is a product of consecutive integers, so one factor is even. Checking $n = 1, 2, 3$ is only three cases; 'if n is even ...' leaves odd n untreated; 'both n^2 and n are even' is false for odd n ; and the contradiction attempt ends with an unresolved exception.

Full method - how and why

What makes a proof **complete**: it must cover every case in the claim with a general argument, not a sample of values.

- 1 The claim is about **every** positive integer, so any valid proof must handle all n at once.

why: Checking particular values can only ever disprove a universal claim, never prove one.

- 2 Factorise: $n^2 + n = n(n + 1)$ - the product of two **consecutive** integers.

why: Factorising turns a statement about a sum into one about a product, where parity is easy to control.

- 3 Of any two consecutive integers, exactly one is even; an even factor makes the product even.

why: This single sentence covers both the case n even and the case n odd - no case-splitting needed.

- 4 That argument is general and complete, so it is the proof.

why: Now examine why the others fail, which is the real content of the question.

Where it goes wrong

- Accepting the check of $n = 1, 2, 3$: three cases out of infinitely many prove nothing.
- Accepting 'if n is even then $n^2 + n$ is even' - true, but it leaves odd n untreated, so the proof is incomplete.
- Accepting 'both n^2 and n are even' - simply false when n is odd.
- Accepting the contradiction attempt that ends 'unless n is even' - a proof cannot end with an unresolved exception.

Takeaway A proof of 'for all n ' must be general. Examples illustrate; only argument proves.

Question 2 PRF-02 · Mathematical proof · spec Prf1

Which of the following is a counterexample to the claim '**if n is a positive integer, then $2^n + 1$ is prime**'?

Answer: D

$2^3 + 1 = 9 = 3 \times 3$ is not prime. ($n = 1, 2, 4, 8$ give 3, 5, 17, 257, all prime.)

Full method - how and why

To disprove a universal claim, find one value where the conclusion fails - so test values until one breaks.

- 1 Evaluate $2^n + 1$ for each candidate: $n = 1 \rightarrow 3$, $n = 2 \rightarrow 5$, $n = 3 \rightarrow 9$, $n = 4 \rightarrow 17$, $n = 8 \rightarrow 257$.

why: Systematic evaluation is faster than trying to be clever here.

- 2 $9 = 3 \times 3$ is **not** prime, so $n = 3$ breaks the claim.

why: That is the counterexample: hypothesis satisfied (3 is a positive integer), conclusion false.

- 3 The others - 3, 5, 17, 257 - are all prime, so none of them refutes anything.

why: (257 is prime: it is not divisible by 2, 3, 5, 7, 11, 13, and $17^2 = 289 > 257$.)

Where it goes wrong

- Assuming a large value like $n = 8$ must fail because the numbers get big - size is not the issue.
- Stopping at the first odd value of $2^n + 1$; every one of them is odd, since 2^n is even.

Takeaway Notice the pattern: $2^n + 1$ can only be prime when n is a power of 2 - and 3 is not, which is exactly why it fails.

Question 3 PRF-03 · Mathematical proof · spec Prf1

Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

- I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.
- II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.
- III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.
- IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

Answer: A

Assume (II), deduce p even (I), deduce q even (IV), reach the contradiction (III).

Full method - how and why

Ordering a proof by contradiction. The logical skeleton is fixed: assume the opposite, deduce consequences, hit the contradiction.

- 1 The assumption must come first: II sets up $\sqrt{2} = \frac{p}{q}$ in lowest terms.
why: Nothing can be deduced before the supposition is made, and 'no common factor' is the condition that will later be contradicted.
- 2 Next I: squaring gives $p^2 = 2q^2$, so p^2 is even, so p is even, $p = 2k$.
why: It depends on II, and it introduces k , which IV uses.
- 3 Then IV: substituting $p = 2k$ gives $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.
why: IV cannot precede I because it uses k , which I defines.
- 4 Finally III: p and q are both even, contradicting 'no common factor'. Hence $\sqrt{2}$ is irrational.
why: The conclusion must come after **both** parities are established. Order: II, I, IV, III.

Where it goes wrong

- Placing III before IV - the contradiction needs q even too, so it cannot come earlier.
- Starting with I, which refers to p and q before they have been introduced.

Takeaway Order a proof by dependency: each line may only use things already defined or deduced.

Question 4 PRF-04 · Mathematical proof · spec Prf1

A student wants to prove that **the square of any odd integer leaves remainder 1 when divided by 8**. Which of the following is the most appropriate first step?

Answer: E

A direct proof: $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$, and $k(k + 1)$ is even, so the square is $8m + 1$. Checking cases is not a proof; assuming the claim is circular.

Full method - how and why

Choosing a proof strategy. For a 'for all odd integers' claim, the standard opening is an algebraic parametrisation.

- 1 Represent an arbitrary odd integer as $2k + 1$, k an integer.

why: This one expression covers **every** odd integer, which is what a universal claim needs.

- 2 (For context, the proof then runs: $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$.)

why: Factorising $4k^2 + 4k$ as $4k(k + 1)$ exposes the consecutive pair.

- 3 $k(k + 1)$ is a product of consecutive integers, hence even, say $2m$, so the square is $8m + 1$.

why: That is remainder 1 on division by 8, as claimed.

Where it goes wrong

- Checking 1, 3, 5, 7, 9 - evidence, not proof.
- Writing the odd integer as $2k$, which is the form of an **even** integer.
- Assuming the claim is true - circular.
- Reaching for contradiction ('assume the remainder is not 1'): legal, but here a direct proof is simpler, and the question asks for the most appropriate step.

Takeaway Universal claims about odd/even numbers start by writing the general form: $2k$ or $2k + 1$.

Question 5 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?

Answer: D

$n^3 - n = (n - 1)n(n + 1)$ is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

Full method - how and why

Proof by exhaustive cases: the cases must cover every integer (sufficient) with none redundant (necessary), and they must be chosen to match the divisor in question.

- 1 The claim is about divisibility by 3, so split by remainder on division by 3: $n = 3k, 3k + 1, 3k + 2$.

why: Every integer has exactly one of these three forms, so the cases are exhaustive and non-overlapping.

- 2 Check each: $n^3 - n = (n - 1)n(n + 1)$. If $n = 3k$ the middle factor is a multiple of 3; if $n = 3k + 1$ then $n - 1 = 3k$; if $n = 3k + 2$ then $n + 1 = 3(k + 1)$.

why: Three consecutive integers always include a multiple of 3 - the case split makes that explicit.

- 3 So the product is divisible by 3 in every case.

why: The proof is complete because the cases cover all integers.

Where it goes wrong

- Splitting into even/odd - that is the wrong modulus and tells you nothing about divisibility by 3.
- Using only $3k + 1$ and $3k + 2$, which omits the multiples of 3 themselves - not exhaustive.
- Splitting by sign or by primality: exhaustive, but useless for this claim.

Takeaway For divisibility by m , split into the m residue classes $mk, mk + 1, \dots, mk + (m - 1)$.

Question 6 PRF-08 · Mathematical proof · spec Prf2

It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?

Answer: D

Contrapositive: if $g(x) \leq 0$ then $f(x) \leq 0$. Since $g(2) = -1 \leq 0$, $f(2) \leq 0$ (it could be 0 or negative).

Full method - how and why

Deduction from an implication. You are given information about the **conclusion**, so the contrapositive is the tool.

1 The given rule is $f(x) > 0 \Rightarrow g(x) > 0$.

why: It lets you conclude about g from f , not the other way round - directly, at least.

2 Its contrapositive is: $g(x) \leq 0 \Rightarrow f(x) \leq 0$, which is equally valid.

why: Negate and swap: $\neg(g > 0)$ is $g \leq 0$, $\neg(f > 0)$ is $f \leq 0$.

3 Apply it at $x = 2$: $g(2) = -1 \leq 0$, so $f(2) \leq 0$.

why: That is all that follows - $f(2)$ could be 0 or negative.

Where it goes wrong

- Concluding $f(2) < 0$ - too strong; the negation of > 0 includes $= 0$.
- Concluding nothing, forgetting that the contrapositive is always available.
- Concluding $f(2) > 0$ by using the implication backwards (affirming the consequent).

Takeaway Information about the conclusion of an implication is used via the contrapositive - and mind the boundary:

$\neg(x > 0)$ is $x \leq 0$.

Question 7 PRF-13 · Mathematical proof · spec Prf1

Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?

Answer: B

Writing the three as $n - 1, n, n + 1$ gives the sum $3n$ for every integer n - general and correct. Checking $1 + 2 + 3, 2 + 3 + 4, 3 + 4 + 5$ is only cases; $n, 2n, 3n$ are not consecutive; and 'any three integers include a multiple of 3' is invalid reasoning (containing a multiple of 3 does not make the sum one).

Full method - how and why

Choosing the right algebraic representation is most of the work. Centre the variables on the middle term.

1 Represent three consecutive integers as $n - 1, n, n + 1$.

why: Centring on n makes the ± 1 cancel; $n, n + 1, n + 2$ also works but gives $3n + 3$, needing one more line.

2 Add: $(n - 1) + n + (n + 1) = 3n$.

why: The -1 and $+1$ cancel.

3 $3n$ is 3 times an integer, hence divisible by 3, for every integer n .

why: General, so it is a proof.

Where it goes wrong

- Using $n, 2n, 3n$ - those are multiples, not consecutive integers.
- Checking three numerical cases.
- Arguing 'any three integers include a multiple of 3' - false for 1, 2, 4, and in any case containing a multiple of 3 would not make the **sum** a multiple of 3.
- The 'mean' argument that ends by demanding the middle integer be a multiple of 3 - it confuses ' $3 \times$ an integer' with ' $3 \times$ a multiple of 3'.

Takeaway Consecutive integers: use $n - 1, n, n + 1$. Their sum is $3n$, always divisible by 3.

Question 8 PRF-14 · Mathematical proof · spec Prf1

To prove 'if n^2 is even then n is even' by contradiction, which assumption should be made?

Answer: B

Assume the hypothesis holds and the conclusion fails, then derive a contradiction: n odd gives n^2 odd, contradicting n^2 even.

Full method - how and why

Setting up a proof by contradiction for an implication: assume the hypothesis **and** the negated conclusion.

- 1 The claim is 'if n^2 is even then n is even': hypothesis n^2 even, conclusion n even.
why: Identify both parts before negating anything.
- 2 For contradiction, suppose the whole implication fails. An implication fails only when the hypothesis holds and the conclusion does not.
why: This is the one case that would make $A \Rightarrow B$ false.
- 3 So assume: n^2 is even **and** n is odd.
why: Both, not just one.
- 4 (The proof then runs: $n = 2k + 1 \Rightarrow n^2 = 4k^2 + 4k + 1$, odd - contradicting n^2 even.)
why: The contradiction is with the retained hypothesis.

Where it goes wrong

- Assuming only ' n is odd' and dropping the hypothesis - then there is nothing to contradict.
- Assuming ' n^2 is odd', which negates the hypothesis instead of the conclusion.
- Assuming ' n even and n^2 odd', which negates the wrong part of each.

Takeaway To contradict $A \Rightarrow B$, assume A **and** $\neg B$ - keep the hypothesis, negate only the conclusion.

Question 9 PRF-16 · Mathematical proof · spec Prf4

A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$** is given with its lines scrambled.

I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.

II Expanding, $a - 2\sqrt{ab} + b \geq 0$.

III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

Answer: A

Start from a known truth (III), expand (II), rearrange (I). Starting from the result and working backwards (I, II, III) would need every step to be reversible and is presented in the wrong direction.

Full method - how and why

Direction matters. A proof should start from something known and deduce the result, not start from the result.

1 Start with a known truth: $(\sqrt{a} - \sqrt{b})^2 \geq 0$, valid because squares are non-negative and $\sqrt{a}, \sqrt{b}$ exist for $a, b > 0$. That is III.

why: Beginning from an undeniable fact is the safe way round.

2 Expand it: $a - 2\sqrt{ab} + b \geq 0$. That is II.

why: $(\sqrt{a})^2 = a$, $(\sqrt{b})^2 = b$, and the cross term is $-2\sqrt{a}\sqrt{b} = -2\sqrt{ab}$.

3 Rearrange: $a + b \geq 2\sqrt{ab}$, then divide by 2. That is I.

why: Dividing by the positive number 2 preserves the inequality.

4 Order: III, II, I.

why: Known truth $\rightarrow$ algebra $\rightarrow$ required result.

Where it goes wrong

- Choosing I, II, III - that starts by assuming the result. Working backwards is a legitimate way to **find** a proof, but to **present** one every step must be reversed.
- Starting with II, whose inequality has not yet been justified.

Takeaway Present proofs forwards, from an established fact to the conclusion. This one proves $AM \geq GM$.

Question 10 PRF-17 · Mathematical proof · spec Prf1

Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

Answer: B

$a = 1, b = -2$: $1 > -2$ but $1 < 4$. The others are true.

Full method - how and why

Scan the claims for the one that squaring breaks - negative numbers are where inequalities misbehave.

- 1 ' $a > b \Rightarrow a^2 > b^2$ ' for all real a, b : try $a = 1, b = -2$. Then $1 > -2$ but $1 < 4$. **False**.
why: Squaring reverses order once negatives with larger magnitude are involved.
- 2 ' $a > b > 0 \Rightarrow a^2 > b^2$ ' is **true** - the positivity condition is exactly what the first claim lacked.
why: Comparing the two shows precisely which hypothesis does the work.
- 3 ' $a > b \Rightarrow a + c > b + c$ ' is **true**: adding the same number to both sides preserves order.
why: Valid for every real c , positive or negative.
- 4 ' $a > b \Rightarrow a^3 > b^3$ ' is **true**: cubing is strictly increasing for every real number, unlike squaring.
why: x^3 keeps the sign of x , so no reversal occurs.
- 5 ' $a > b \Rightarrow -a < -b$ ' is **true**: multiplying by -1 reverses the inequality.
why: The standard sign rule.

Where it goes wrong

- Thinking the cubing claim fails - it does not; only **even** powers destroy order.
- Thinking multiplying by -1 keeps the direction.

Takeaway Squaring preserves order only for non-negative numbers; cubing, adding and negating (with a flip) always behave.

Question 11 PRF-20 · Mathematical proof · spec Prf3

T_n denotes the n th triangular number $\frac{n(n+1)}{2}$. A student observes $T_1 + T_2 = 4$, $T_2 + T_3 = 9$, $T_3 + T_4 = 16$ and conjectures that $T_n + T_{n+1}$ is always a perfect square. Which of the following proves the conjecture?

Answer: A

Only the algebraic verification $T_n + T_{n+1} = \frac{(n+1)(2n+2)}{2} = (n+1)^2$ is general; the rest check one more case, assert something false, or state a true identity without showing the result.

Full method - how and why

Proving a conjecture about a sequence: substitute the formula and simplify algebraically to a general result.

- 1 Write both terms: $T_n = \frac{n(n+1)}{2}$ and $T_{n+1} = \frac{(n+1)(n+2)}{2}$.
why: Replace n by $n+1$ throughout the given formula.
- 2 Add over the common denominator: $\frac{n(n+1) + (n+1)(n+2)}{2}$.
why: Same denominator, so just add numerators.
- 3 Factor out $(n+1)$: $\frac{(n+1)[n + (n+2)]}{2} = \frac{(n+1)(2n+2)}{2}$.
why: Spotting the common factor avoids expanding and re-factorising.
- 4 $2n+2 = 2(n+1)$, so the expression is $(n+1)^2$ - a perfect square for every n . **Proved**.
why: The 2 cancels the denominator.

Where it goes wrong

- Checking $T_4 + T_5 = 25$ and stopping - one more case is still not a proof.
- 'Triangular numbers are always squares' - plainly false ($T_2 = 3$).
- ' $T_n + T_{n+1} = 2T_n + (n+1)$, which is a square' - the identity is right but the conclusion is asserted, not shown.
- Quoting the difference $T_{n+1} - T_n = n+1$, which is true but says nothing about the sum.

Takeaway Substitute the general formula and simplify; consecutive triangular numbers sum to $(n+1)^2$.

Question 12 X2-27 · Mathematical proof · spec Prf1

Consider the claim: **if n is a positive integer then $n^2 + n + 1$ is odd**. Which of the following is a valid proof?

Answer: A

The argument from $n^2 + n = n(n + 1)$ being even is direct and general. Checking $n = 1, 2, 3$ is only cases; 'cannot be factorised' is irrelevant to parity; 'if n is odd ...' says nothing about even n ; and the contradiction attempt never reaches a contradiction, since n may well be even.

Full method - how and why

Same 'which is a valid proof' pattern as PRF-01: general argument versus cases, irrelevance and incompleteness.

- 1 The valid proof: $n^2 + n = n(n + 1)$ is a product of consecutive integers, hence even; adding 1 gives an odd number.
why: General, covering every positive integer with no case split.
- 2 Checking $n = 1, 2, 3$ gives 3, 7, 13 - evidence, not proof.
why: Three cases out of infinitely many.
- 3 'It cannot be factorised, so it is odd' - factorisability has nothing to do with parity ($6 = 2 \times 3$ and $9 = 3 \times 3$ are respectively even and odd).
why: An irrelevant premise cannot support the conclusion.
- 4 'If n is odd then ... so $n^2 + n + 1$ is odd' - correct but only for odd n ; even n is untreated. Incomplete.
why: A case-based proof must cover every case.
- 5 'Suppose $n^2 + n + 1$ is even. Then n is even, contradiction' - n being even is not a contradiction; n can be even.
why: A contradiction proof must actually reach an impossibility.

Where it goes wrong

- Accepting the odd- n argument because every line in it is true - truth is not completeness.
- Accepting the contradiction attempt because it uses the right vocabulary.

Takeaway $n(n + 1)$ is always even, so $n^2 + n + 1$ is always odd. A proof must be general, relevant and complete.